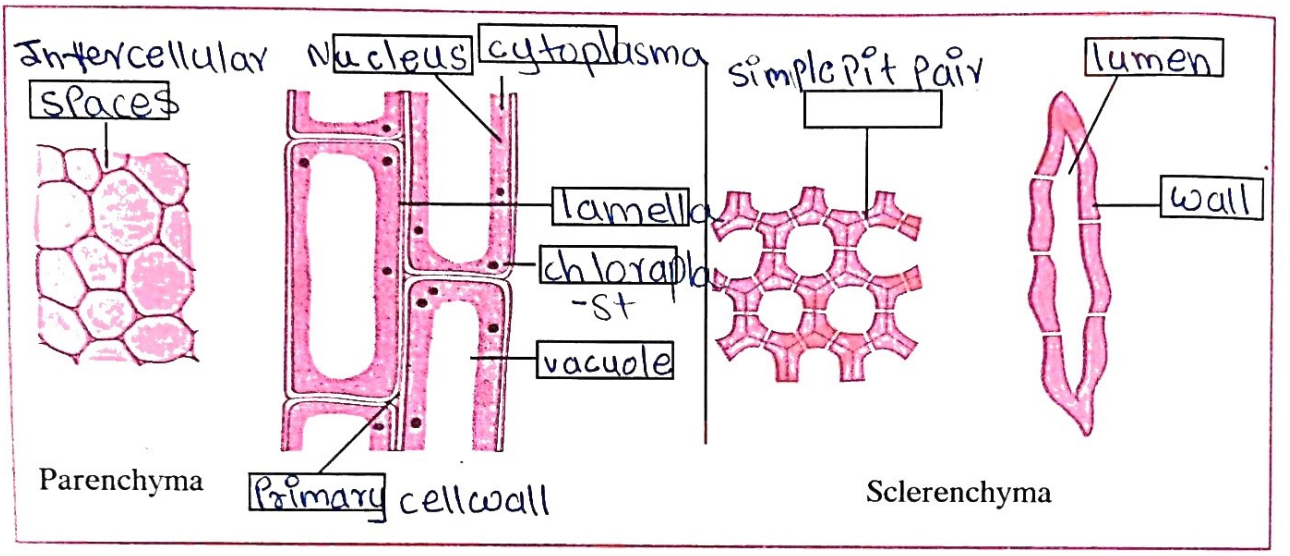
Experiment No. 13

Title : Identify the parenchyma and sclerenchyma tissues in plants from the permenent slides and draw neat and labelled diagrams.

Apparatus : Permenent slides and charts of parenchyma and sclerenchyma, Compound microscope, etc.

Figure :



Procedure :

1. Observe the permanent slides of parenchyma and sclerenchyma tissues under high magnification of compound microscope.
2. Record the observations in the table and draw their well labelled diagrams.

Observation table :

Features	Parenchyma	Sclerenchyma
Cell wall	Thin wall cell	Thick wall cell
Staining of cell wall	made up of calulose	mainly formed lignin
Arrangement of cells	Arranged longitudinally	Stone like cell
Intracellular spaces	Present between cell	Absent
Shape of cells	oval, Polygonal shape	long and Narrow
Lumen or cell cavity	Present	Absent
Nucleus	Present	Absent (Dead tissue)
Living / Dead	living cell.	dead cell
Occurance	Packing tissue	Out hypodermis
Functions	1) Give support 2) Storage of food	Gives strength & rigidity.

Multiple Choice Questions

1. Epidermis of stem and leaves have.....layer.
   A. pectin B. suberin C. cuticle D. lignin
2. Presence of aerenchyma tissue in stem and leaves is a characteristics of .....plants.
   A. aquatic B. desert/terrestrial C. perenial D. climber
3. ..... and .....are the types of permanent tissues.
   A. Collenchyma and Sclerenchyma B. Chlorenchyma and Aerenchyma
   C. Meristmatic and Surface tissue D. Chlorenchyma and Sclerenchyma
4. The walls of sclerenchyma tisssue are thicker due to the layer of .....
   A. pectin B. suberin C. cellulose D. lignin
5. .....tissues are present in the leaf petiole.
   A. Parenchyma B. Collenchyma C. Sclerenchyma D. Surface

Questions

1. Explain the types of parenchyma tissues with their functions.

1) It is the type of simple permanent tissue.
2) Types of parenchyma - a) chlorenchyma
b) Aerenchyma.
a) function of chlorenchyma :- Leaves perform Photosyn-
-thesis.
b) function of Aerenchyma :- Helps aquatic plants, leaves
and stem to flaat.

2. How sclerenchyma tissues are important for plant though they are dead tissue ?

1) In sclerenchyma cell wall contain lignin.
2) It gives strength and rigidity to parts of the plant.
Therefore sclerenchyma tissue are important for
plant though they are ~~ded~~ dead tissue. Sclerenchyma.

3. In which type of living organisms meristmatic tissues are found ? State their importance.

meristmatic tissue found in plants.

Importance :- There are same concentrated
spores in plants that are not discriminated
and assist in the germination of the
plants.

H88JXT